Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.001 with the Adam optimizer. The dataset was split into 70 percent training data and the rest for testing.

2 Results

Figure 1 shows the f1 score of Ours by dataset.

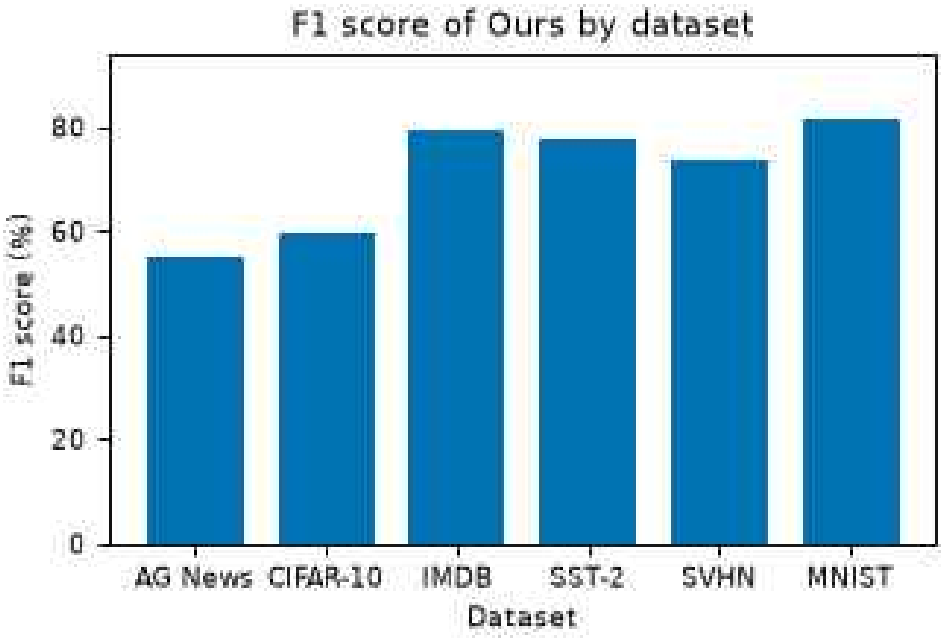


Figure 1: F1 score of Ours by dataset for the models in this experiment.

Ours reaches 81.6% F1 score on MNIST. Ours scores 2.0 points higher on MNIST than on IMDB.

Figure 2 shows the f1 score by dataset.

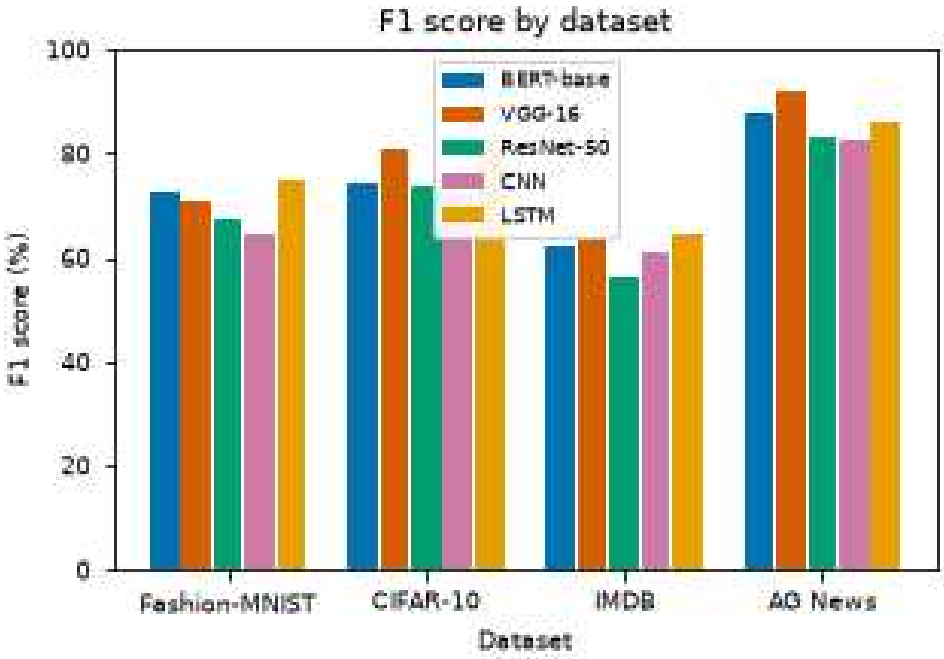


Figure 2: F1 score by dataset for the models in this experiment.

LSTM outperforms ResNet-50 by about 7 percentage points on Fashion-MNIST (Figure 2). LSTM reaches

54.5% F1 score on IMDB.

Figure 3 shows the accuracy during training.

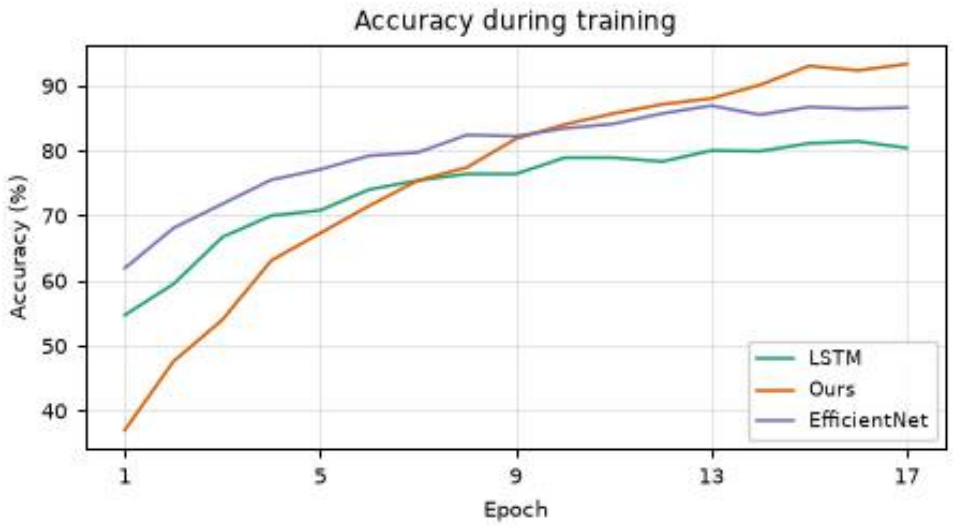


Figure 3: Accuracy during training for the models in this experiment.

As shown in Figure 3, from epoch 9 to epoch 10, the accuracy of LSTM improves by a relative 11.4%. At epoch 15, LSTM achieves over 80% accuracy (Figure 3). EfficientNet attains a accuracy of about 90% at epoch 15. As shown in Figure 3, from epoch 8 to epoch 13, the accuracy of Ours improves by a relative 29.7%.

3 Discussion

Random Forest reaches 53.5% accuracy on CIFAR-10. Overall, the results are in line with what we expected from the literature.